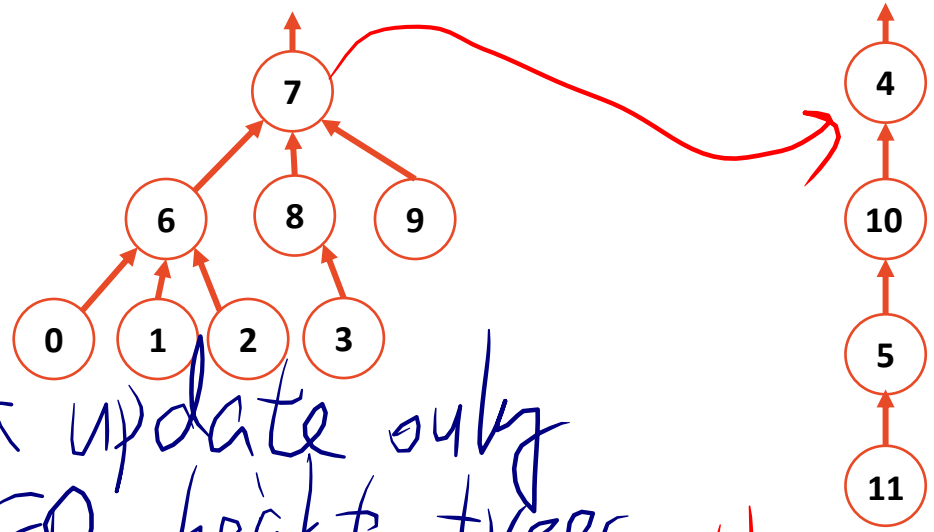
CS 225

Data Structures

Volodymyr Kindratenko

Slides courtesy of Wade Fagen-Ulmschneider

Disjoint Sets – Smart Union



Union by height

0	1	2	3	4	5	6	7	8	9	10	11
6	6	6	8	-4	10	7		7	7	4	5

***Idea**: Keep the height of the tree as small as possible.*

Union by size

0	1	2	3	4	5	6	7	8	9	10	11
6	6	6	8		10	7		7	7	4	5

***Idea**: Minimize the number of nodes that increase in height*

Both guarantee the height of the tree is: _____.

Disjoint Sets Find

Find runs in O(log n)

```
1  int DisjointSets::find(int i) {
2    if ( s[i] < 0 ) { return i; }
3    else { return find( s[i] ); }
4  }
```

```
1  void DisjointSets::unionBySize(int root1, int root2) {
2    int newSize = arr_[root1] + arr_[root2];
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4    // If arr_[root1] is less than (more negative), it is the larger set;
5    // we union the smaller set, root2, with root1.
6    if ( arr_[root1] < arr_[root2] ) {
7      arr_[root2] = root1;
8      arr_[root1] = newSize;
9    }
10
11   // Otherwise, do the opposite:
12   else {
13     arr_[root1] = root2;
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15   }
16 }
```
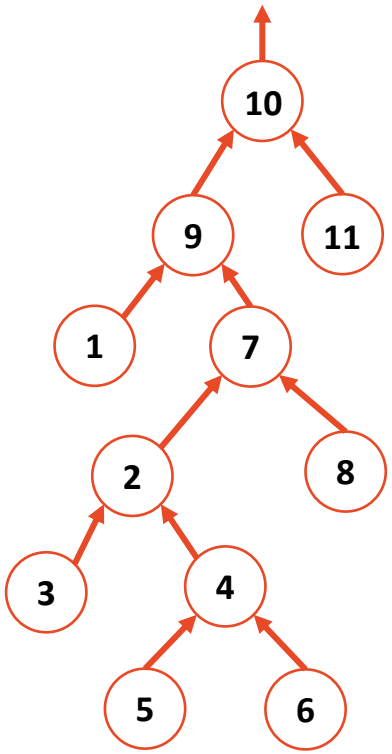

-5 + -4

Given two roots O(1)

Path Compression



find(4)

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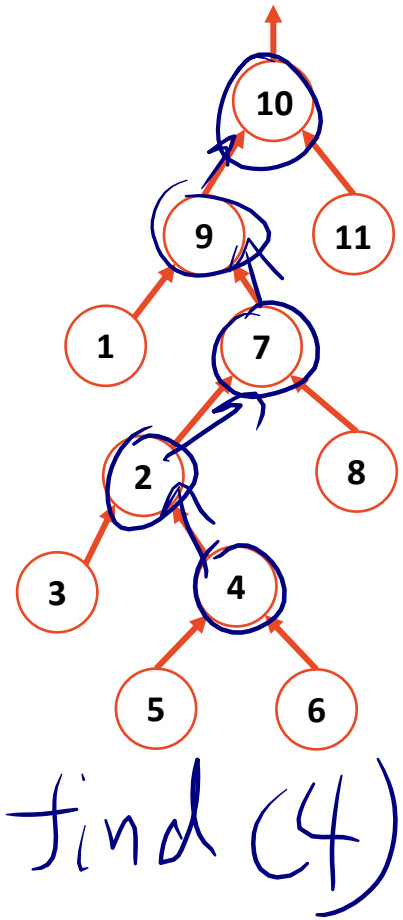

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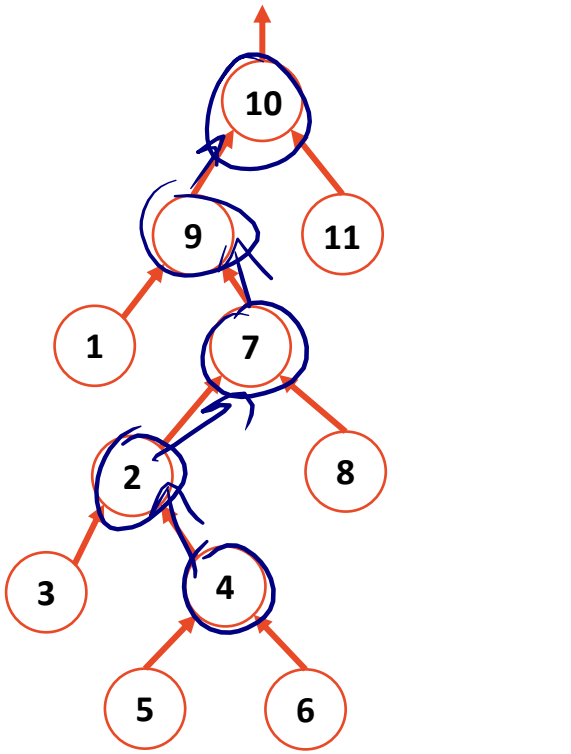

find(4) = 10

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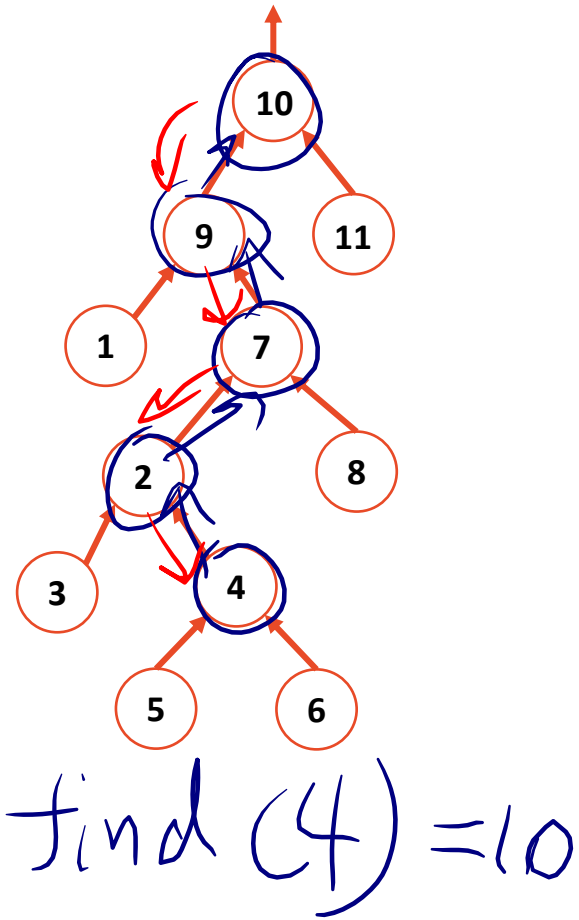

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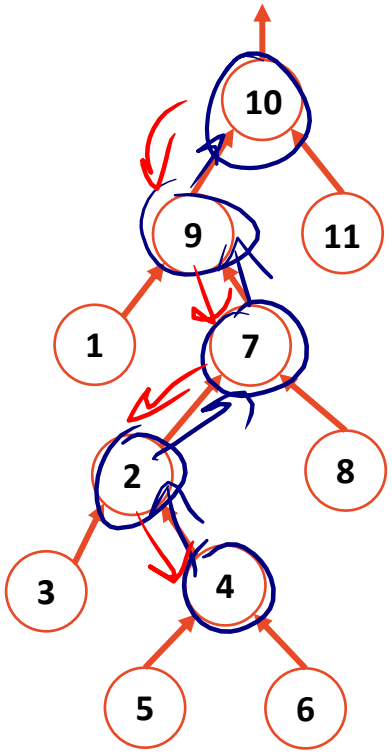

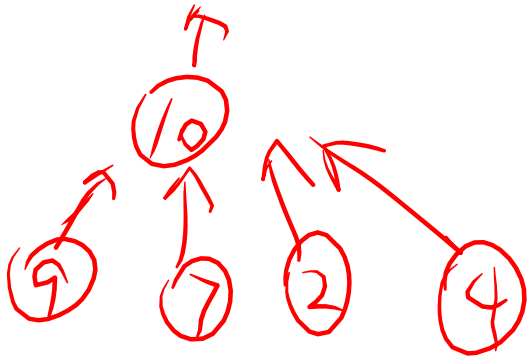

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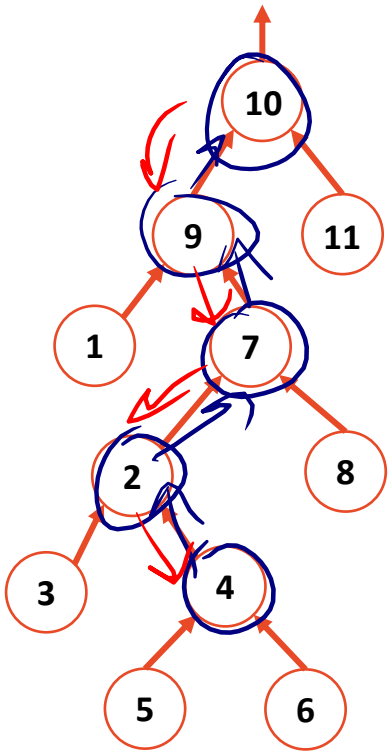

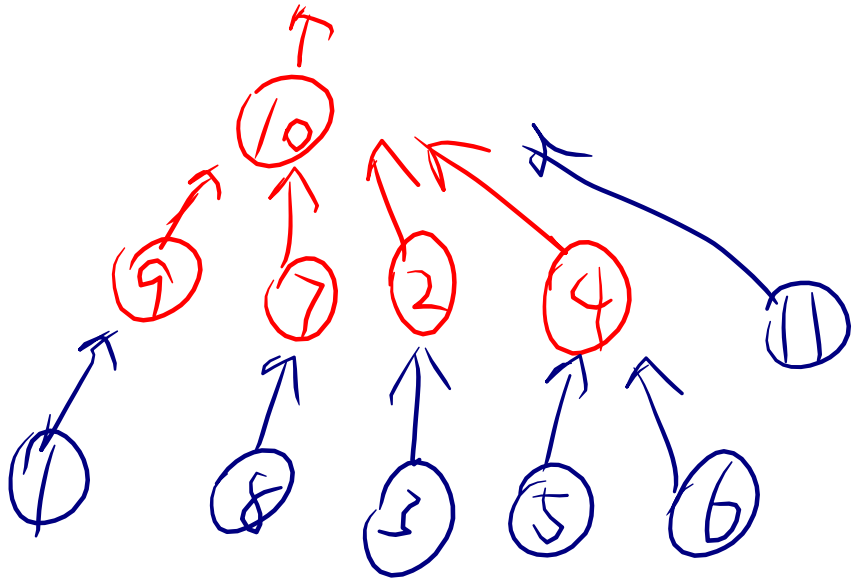

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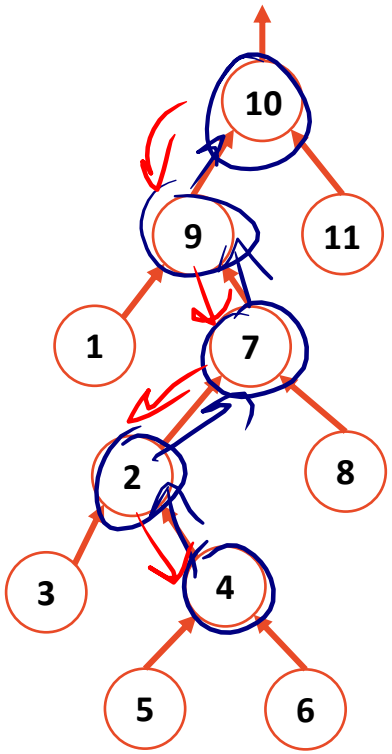

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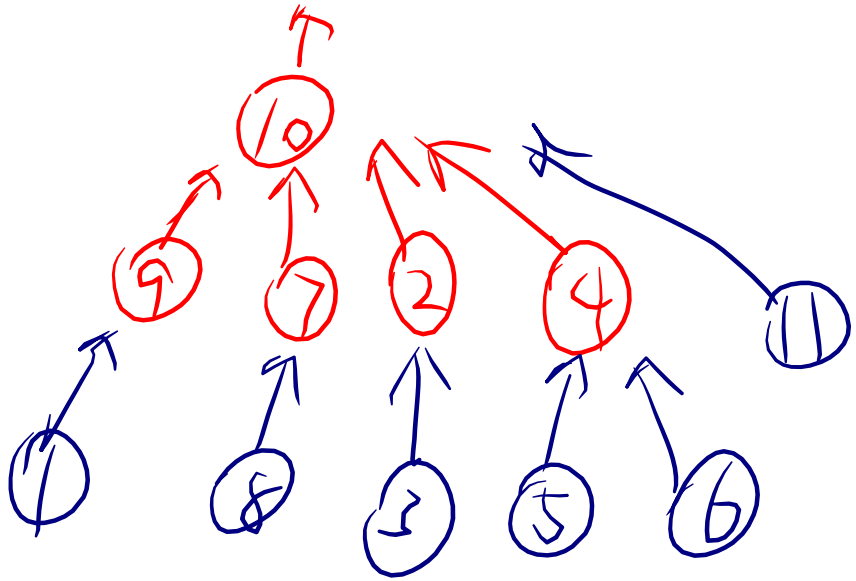

Ideal O(1)

Disjoint Sets Analysis

The **iterated log** function:

The number of times you can take a log of a number.

$\log^*(n) =$

$$\begin{cases} 0 & , n \leq 1 \\ 1 + \log^*(\log(n)) & , n > 1 \end{cases}$$

What is $\mathbf{\log^*(2^{65536})}$?

↳ 65536

(1)

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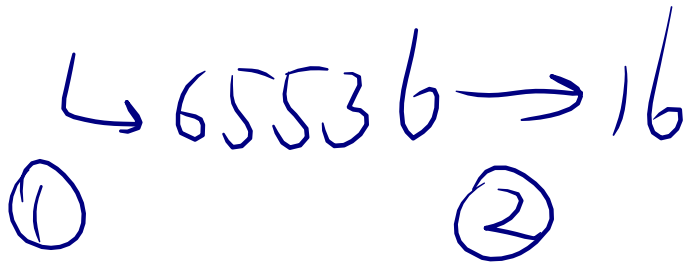

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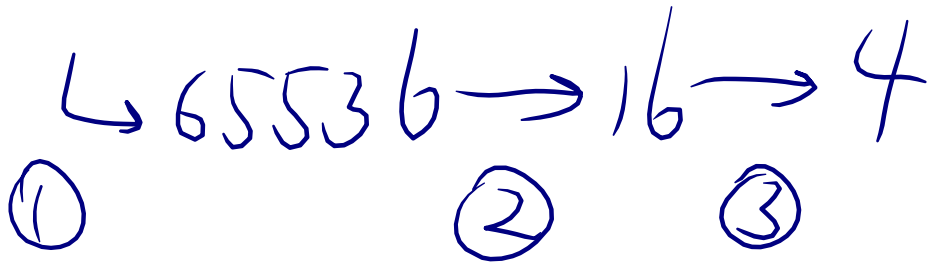

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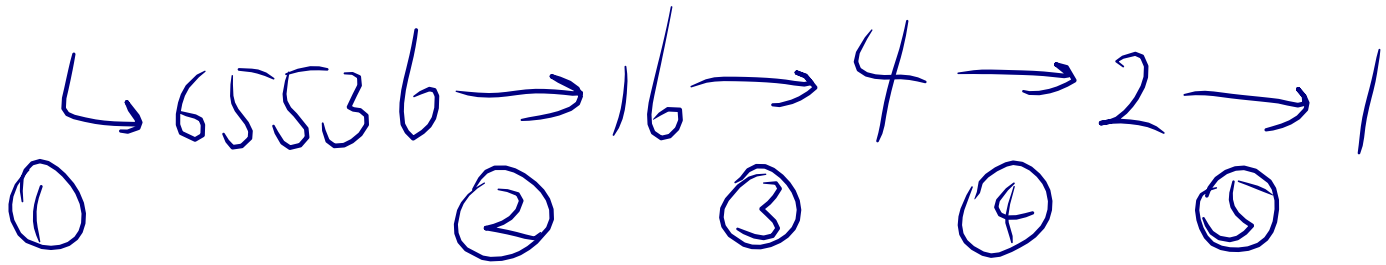

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What is $\log^*(2^{65536})$? $= 5$

$\hookrightarrow 65536 \rightarrow 16 \rightarrow 4 \rightarrow 2 \rightarrow 1$

① ② ③ ④ ⑤

Disjoint Sets Analysis

In an Disjoint Sets implemented with smart **unions** and path compression on **find**:

Any sequence of **m union** and **find** operations result in the worse case running time of O(_____),

where **n** is the number of items in the Disjoint Sets.

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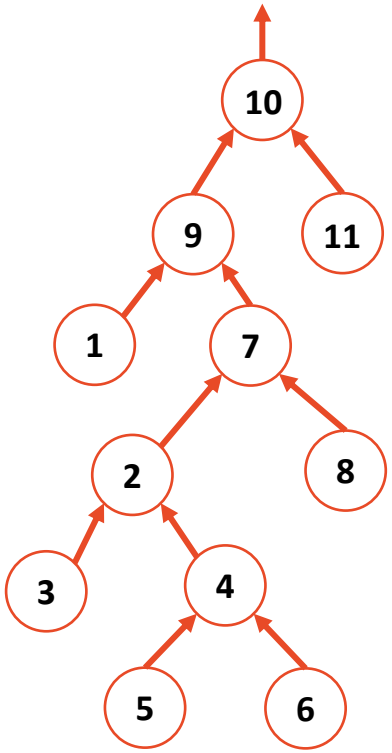
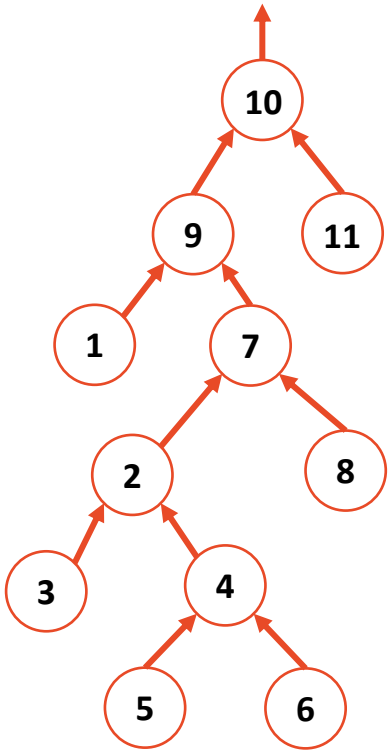
UpTree



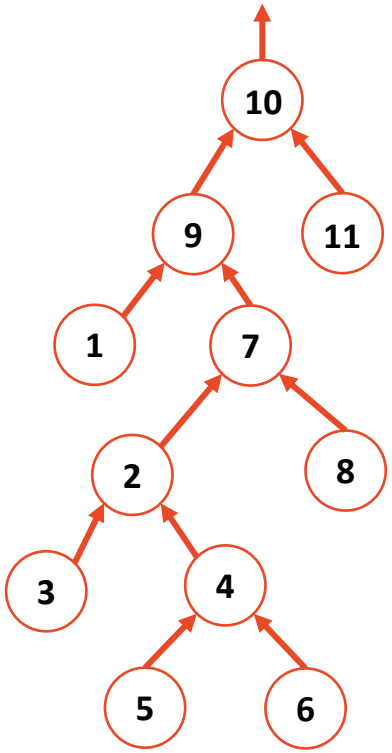
UpTree

Uptree - Impl w/ array



UpTree

Uptree - Impl w/ array
- Reps. a set



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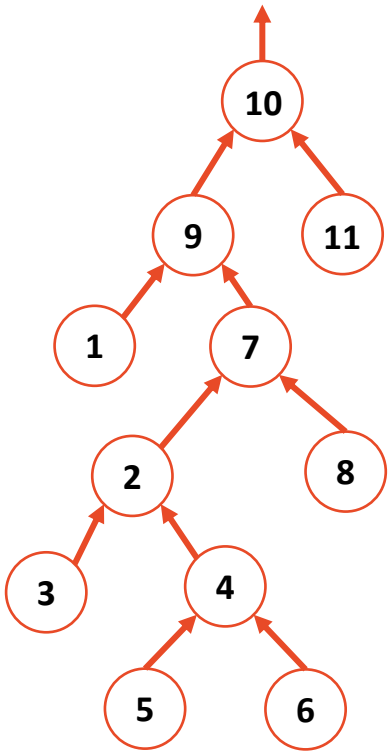
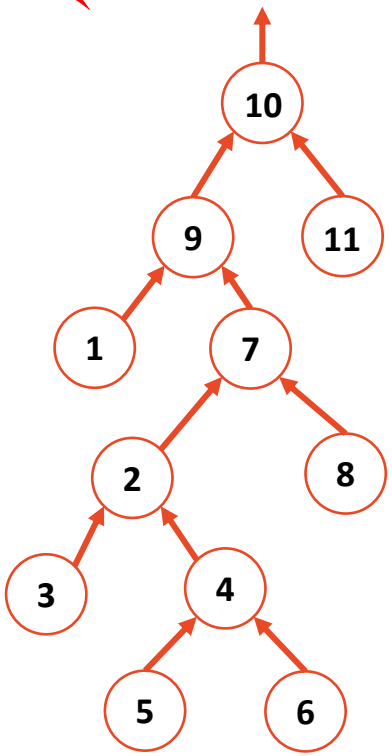


Uptree - Impl w/ array
- Reps. a set

A forest of uptrees rep. a collection of EQ. classes (disjoint sets)

UpTree

[10]R



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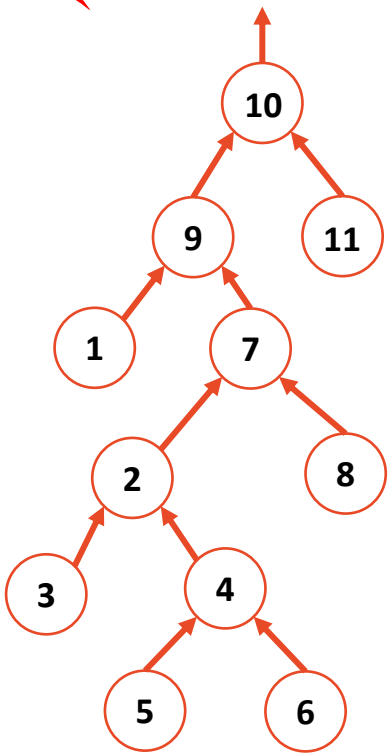

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Near O(1) to find rep. elem.

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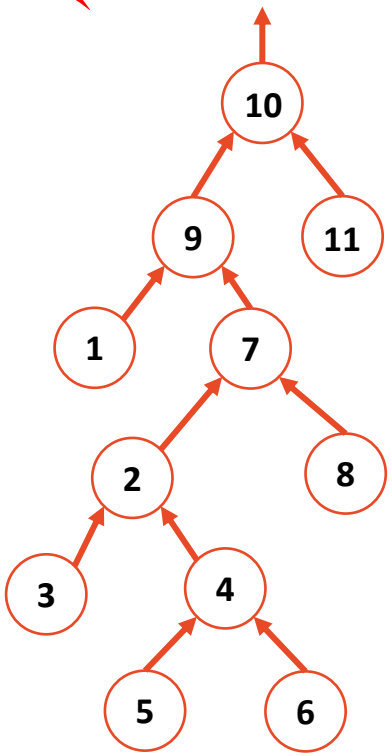

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Near O(1) to find rep. elem.

Cannot quickly find all elem. in a set

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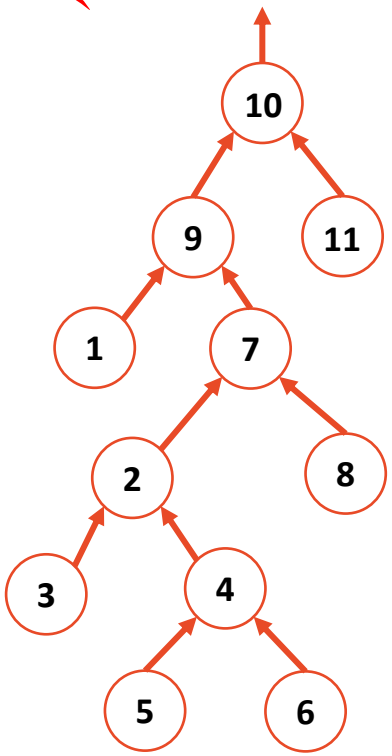

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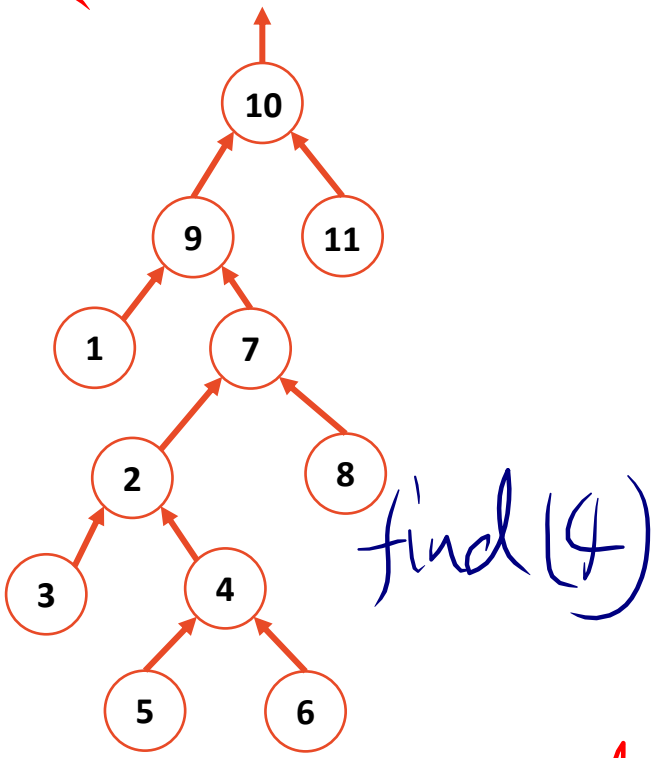

find(4)

☆Near O(1) to find rep. elem.

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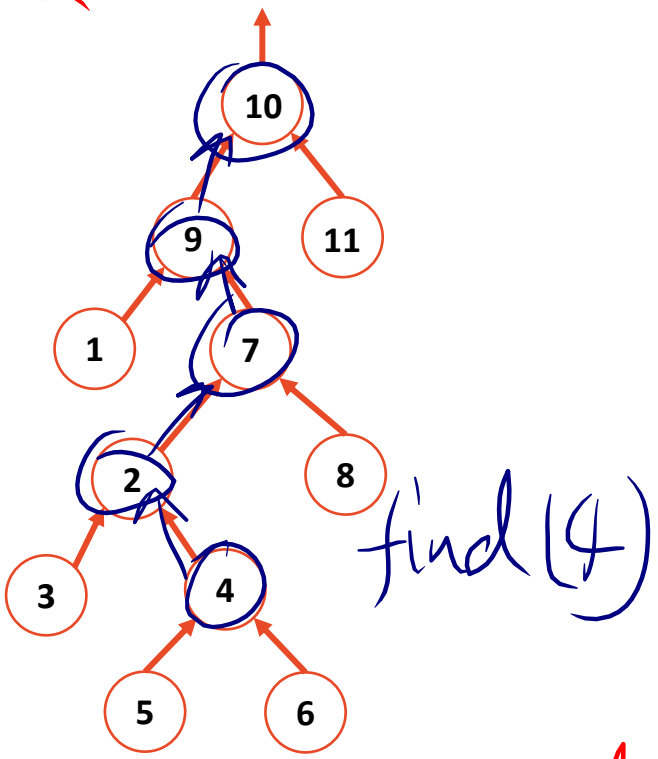

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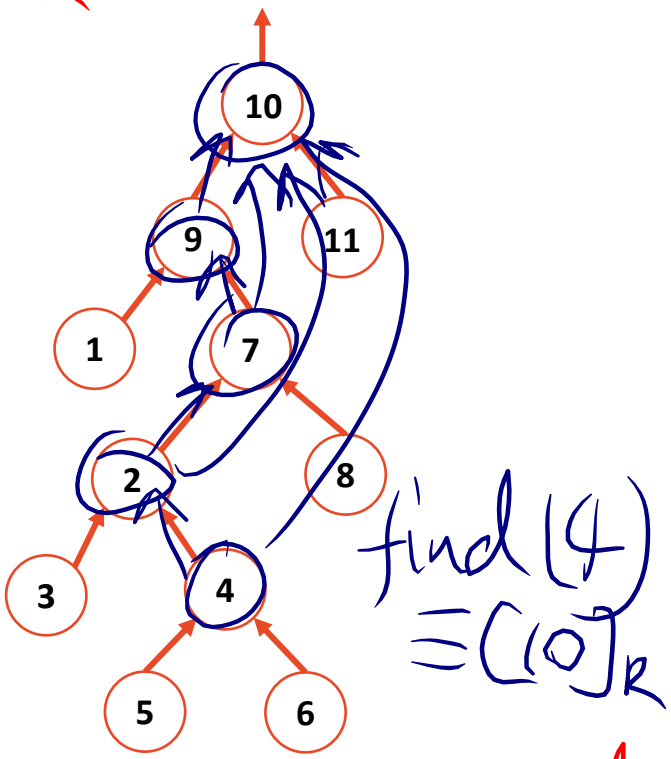

find(4) $= [10]_R$

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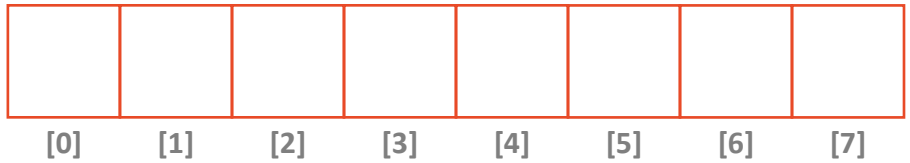
In Review: Data Structures

Array

- Sorted Array
- Unsorted Array
  - Stacks
  - Queues
  - Hashing
  - Heaps
    - Priority Queues
  - UpTrees
    - Disjoint Sets

List

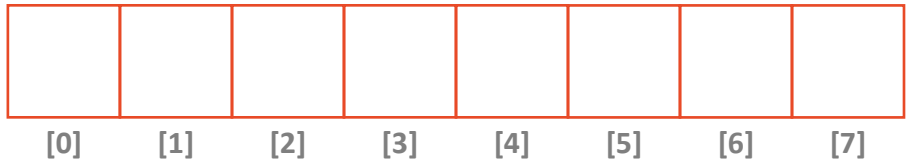
- Doubly Linked List
- Skip List
- Trees
  - BTree
  - Binary Tree
    - Huffman Encoding
    - kd-Tree
    - AVL Tree



Array

- Constant time access to any element, given an index
  a[k] is accessed in O(1) time, no matter how large the array grows

- Cache-optimized
  Many modern systems cache or pre-fetch nearby memory values due the “Principle of Locality”. Therefore, arrays often perform faster than lists in identical operations.



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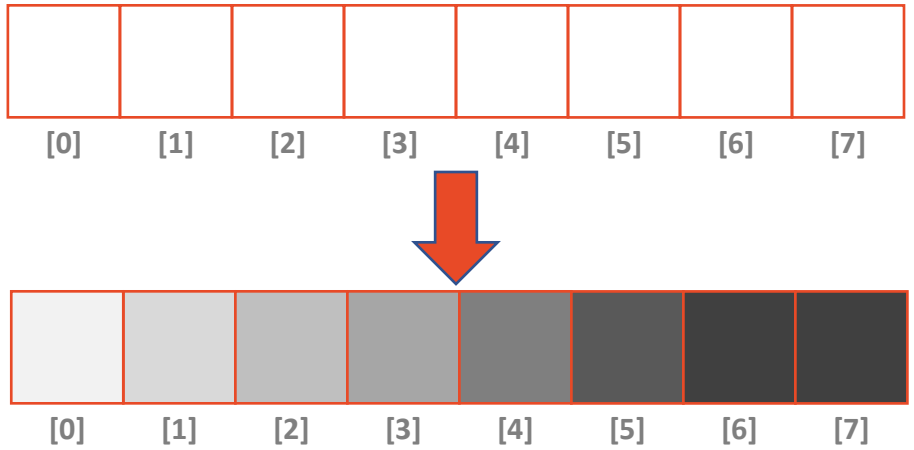


Array

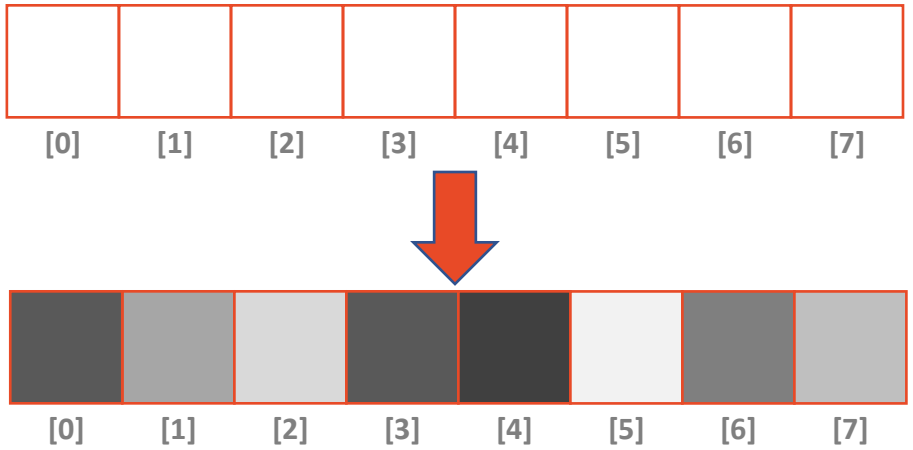
Sorted Array

- Efficient general search structure

  Searches on the sort property run in O(log(n)) with Binary Search

- Inefficient insert/remove

  Elements must be inserted and removed at the location dictated by the sort property, resulting shifting the array in memory – an O(n) operation

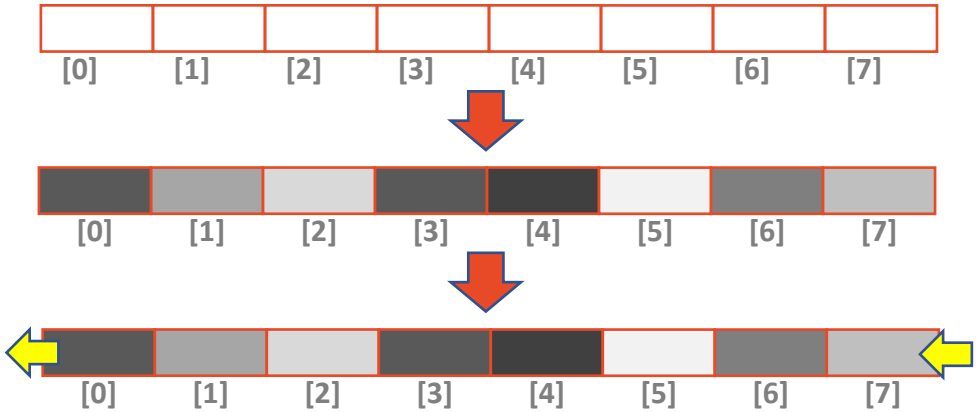


Array

Unsorted Array

- Constant time add/remove at the beginning/end
  Amortized O(1) insert and remove from the front and of the array
  <u>Idea:</u> Double on resize

- Inefficient search structure
  With no sort property, all searches must iterate the entire array; O(n) time



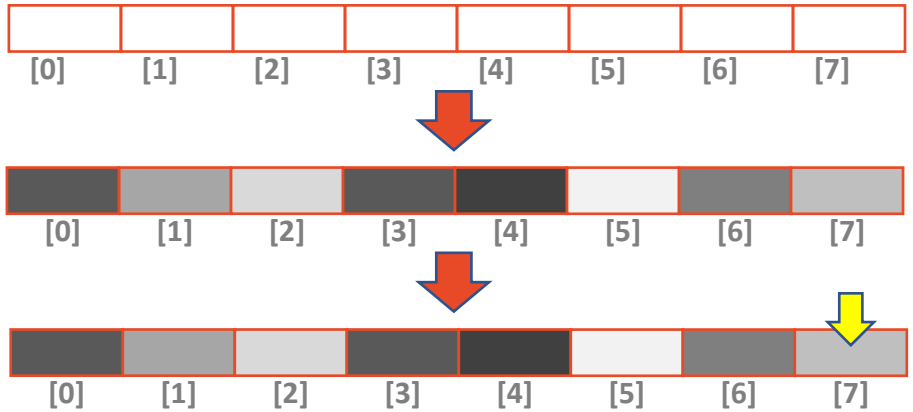
Array

Unsorted Array

Queue (FIFO)

- First In First Out (FIFO) ordering of data
  Maintains an arrival ordering of tasks, jobs, or data

- All ADT operations are constant time operations
  enqueue() and dequeue() both run in O(1) time



Array

Unsorted Array

Stack (LIFO)

- Last In First Out (LIFO) ordering of data
  Maintains a "most recently added" list of data

- All ADT operations are constant time operations
  push() and pop() both run in O(1) time

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- Skip List
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